ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** curiosity	A curious mind keeps asking why and how.
2	**(a)** a systematic way of understanding nature	Science studies nature through observation and experiment.
3	**(b)** a mobile phone	Technology is the practical use of science.
4	**(a)** observation	Observation is the first step of enquiry.
5	**(a)** a mango tree	Plants grow, respire and reproduce.
6	**(d)** technology	Technology applies science.
7	**(a)** hypothesis	A hypothesis is a testable explanation.
8	**(d)** make sure the result is reliable	Repetition checks reliability.
9	**(b)** vaccines that prevent diseases	Vaccines were developed by science.
10	**(c)** recording and analysing results	Results are recorded and studied before a conclusion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Repetition checks that the result is reliable and not due to chance (1); It reduces the effect of errors (1).	1+1
12	Science: understanding how nature works (1); Technology: using that knowledge to make useful things (1).	1+1
13	Mobile phone or refrigerator (1); Electric fan or pressure cooker (1).	1+1
14	An intelligent guess that can be tested (1); Example: plants kept in sunlight grow taller than plants kept in the dark (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Observe and ask a question (1); Make a hypothesis and test it by experiment (1); Record results and draw a conclusion (1).	1+1+1
16	Observation: what we see, e.g. the ground is wet (1); Inference: what we conclude, e.g. it has rained (1); Inference should be supported by evidence (1).	1+1+1
17	Observation: watching things as they are (1); Experiment: changing a condition to see the effect (1); An experiment is done to test a hypothesis (1).	1+1+1
	OR: Hypothesis: plants grow better in sunlight (1); Keep two identical plants, one in light and one in dark (1); Give same water and soil to both (1).	1+1+1
18	Plants need water to live (1); He made an observation (1); Conclusion: water is necessary for plant growth (1).	1+1+1
	OR: It magnifies very small objects (1); Cells or tiny organisms can be seen (1); They are not visible to the naked eye (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Observation: dry seeds do not sprout (1); Hypothesis: seeds need water (1); Experiment: one dish with water, one dry, same seeds and place (1); Result: only wet seeds sprout (1); Conclusion: seeds need water for germination (1).	1+1+1+1+1
	OR: Physics: matter, energy and forces (1); Chemistry: substances and changes (1); Biology: living things (1); Astronomy: space objects (1); Geology: Earth and rocks (1).	1+1+1+1+1
20	Science is the study of nature through observation and tests (1); Technology uses scientific knowledge to make useful things (1); Science: laws of motion, properties of light (1); Technology: bicycle, telescope (1); Both depend on each other (1).	1+1+1+1+1
	OR: Records help recall results later (1); They allow others to check the work (1); Include date and conditions (1); Include measurements with units (1); Include the conclusion (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a guess a **conclusion**: the guess before the experiment is a hypothesis; the conclusion comes after the results.

- Changing many things in an experiment at once; only **one** condition should be changed.
- Writing only what you expected to see instead of what you actually observed.
- Believing that one trial is enough; experiments are repeated to be sure.
- Mixing up the instruments: microscope is for tiny things, telescope is for faraway things.
- Saying science facts never change; new evidence can improve our understanding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Does a seed sprout faster in warm water?	Only measurable questions can be tested.
2	**(b)** botanist	Botany is the study of plants.
3	**(a)** scientific method	It uses observation, hypothesis, experiment and conclusion.
4	**(d)** communicating results	Sharing findings is part of science.
5	**(c)** zoologist	Zoology deals with animals.
6	**(a)** accepting any claim without checking	Scientists test claims before accepting them.
7	**(d)** Aryabhata	Aryabhata lived in the fifth-sixth century CE.
8	**(a)** astronomy	Astronomers observe celestial bodies.
9	**(a)** experiment	An experiment tests a hypothesis.
10	**(d)** surgery	Sushruta Samhita is a text on surgery.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Only one factor is changed at a time (1); All other conditions are kept the same (1).	1+1
12	It gives correct data to build a hypothesis (1); It helps spot patterns and changes (1).	1+1
13	Observation (1); Hypothesis, experiment or conclusion (1).	1+1
14	Telescope for the sky (1); Microscope for tiny things (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Others can learn from the findings (1); They can repeat the experiment to verify it (1); It allows science to grow by building on earlier work (1).	1+1+1
16	Follow the teacher's instructions (1); Do not taste or smell chemicals directly (1); Wash hands after the activity (1).	1+1+1
17	Curiosity leads to questions (1); Questions lead to experiments and discoveries (1); Example: wondering why apples fall led to studying gravity (1).	1+1+1
	OR: Botany: plants (1); Zoology: animals (1); Astronomy: stars, planets and space (1).	1+1+1
18	He studied how light scatters when passing through substances (1); This is known as the Raman effect (1); He received the Nobel Prize in Physics in 1930 (1).	1+1+1
	OR: Conclusion is a statement based on results (1); Hypothesis is a guess made before testing (1); Conclusion can accept or reject the hypothesis (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Curiosity drives questions (1); Honesty ensures true records (1); Patience is needed for repeated experiments (1); Example: Edison tested many materials for a bulb filament (1); These habits make results trustworthy (1).	1+1+1+1+1
	OR: Electricity for lighting (1); Medicines and vaccines (1); Transport such as trains and aeroplanes (1); Mobile phones and the internet (1); Refrigeration to store food (1).	1+1+1+1+1
20	A fair test changes only one variable (1); Balls of same size, different mass (1); Drop from same height at the same time (1); Observe which lands first (1); Repeat to confirm the result (1).	1+1+1+1+1
	OR: Observation: plant bends towards window (1); Hypothesis: plants grow towards light (1); Test: rotate the pot and watch (1); Result: the plant bends again towards light (1); Conclusion: plants respond to light (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a guess a **conclusion**: the guess before the experiment is a hypothesis; the conclusion comes after the results.
- Changing many things in an experiment at once; only **one** condition should be changed.
- Writing only what you expected to see instead of what you actually observed.
- Believing that one trial is enough; experiments are repeated to be sure.

- Mixing up the instruments: microscope is for tiny things, telescope is for faraway things.
- Saying science facts never change; new evidence can improve our understanding.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** geology	Geologists study the Earth.
2	**(a)** microscope	A microscope magnifies tiny objects.
3	**(b)** scattering of light	The Raman effect concerns scattering of light.
4	**(c)** chemistry	Chemistry deals with matter and its changes.
5	**(b)** telescope	Telescopes make faraway objects appear closer.
6	**(b)** using proper instruments and careful reading	Accurate tools and careful reading give reliable data.
7	**(b)** wearing proper gear and following instructions	Safety rules prevent accidents.
8	**(d)** doing activities and experiments	Hands-on activity builds understanding.
9	**(c)** changing only one thing at a time	Only one variable should change.
10	**(c)** the results obtained	Conclusions follow from evidence.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	An ancient Indian scholar (1); He worked in mathematics and astronomy (1).	1+1
12	It makes us ask questions about what we see (1); Questions lead to investigations and new discoveries (1).	1+1
13	Others can check and repeat the work (1); False records lead to wrong conclusions (1).	1+1
14	They may be poisonous or harmful (1); Labs follow rules to prevent accidents (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mobile phones for communication (1); Refrigerators to preserve food (1); Vaccines and medicines for health (1).	1+1+1
16	By observing nature (1); By doing experiments (1); By reading and discussing with others (1).	1+1+1
17	So that results can be compared (1); Changing conditions would change results (1); It tells us if the result is consistent (1).	1+1+1
	OR: Curiosity (1); Honesty in recording data (1); Willingness to repeat and check results (1).	1+1+1
18	Measurement gives a definite value in units (1); Values can be compared and checked (1); Guesses differ from person to person (1).	1+1+1
	OR: Aryabhata: mathematics and astronomy (1); C.V. Raman: physics (light) (1); Sushruta: surgery (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Aryabhata: astronomy and maths (1); Charaka: medicine (Charaka Samhita) (1); Sushruta: surgery (1); They used observation and reasoning (1); Their work still inspires modern science (1).	1+1+1+1+1
	OR: To prevent injury and accidents (1); Wear protective gear (1); Never taste chemicals (1); Handle glass and flames carefully (1); Report spills to the teacher (1).	1+1+1+1+1
20	Microscope: tiny things (1); Telescope: distant objects (1); Thermometer: temperature (1); Balance: mass (1); Measuring tape or scale: length (1).	1+1+1+1+1
	OR: New evidence can change earlier explanations (1); Example: people once thought the Earth was flat (1); Example: people once believed that the Sun moves around the Earth (1); Scientists keep questioning ideas (1); Knowledge grows with more observations (1).	1+1+1+1+1

Mistakes to avoid:

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- Changing many things in an experiment at once; only **one** condition should be changed.
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- Mixing up the instruments: microscope is for tiny things, telescope is for faraway things.

- Saying science facts never change; new evidence can improve our understanding.

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